Kasi - a South African slang term (from township slang, especially Tsotsi taal) that means township.

Kasi Scout.







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Kasi Scout.

KasiScout is a data-driven football scouting platform that discovers and evaluates overlooked players in kasi communities, using video analysis and machine learning to provide insights, predict potential, and provide data-driven insights to support talent identification and recruitment decisions.

What's the problem?

- Many talented players in kasi lack exposure.
- Players have limited access to coaching and resources.
- Traditional scouting misses hidden or underrated talent.

This results in...

- **Talent wasted**: Skilled players never get noticed or opportunities to grow.
- **Inequality in football pathways**: Only players with exposure, money, or connections progress.
- **Missed opportunities for clubs**: Scouts and teams miss hidden talent that could improve their squads.

Who benefits from KasiScout ?

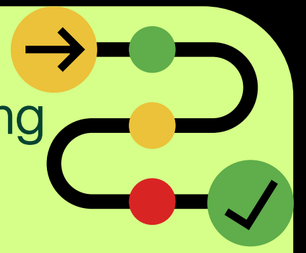


- PLAYERS
- CLUBS/SCOUTS
- FOOTBALL ECOSYSTEM
- COMMUNITIES

Where is the opportunity?

- No existing system effectively captures and evaluates hidden talent in kasi communities.
- Traditional scouting focuses on elite players or tournaments, leaving many skilled players unseen.
- Data and AI make it possible to systematically discover, track, and evaluate talent was previously invisible.

How KasiScout will operate :



1. **Capture Player Activity:**
   - Coaches, local volunteers, or even the players themselves record matches and training sessions using smartphones or affordable cameras.
   - The system collects video footage and basic player info (name, age, position).
2. **Analyze Player Performance:**
   - The software processes videos to track player movements, ball interactions, and actions like passing, shooting, dribbling, and positioning.
   - Generates statistics and visualizations for each player.
3. **Evaluate and Rank Players:**
   - Scores players on performance metrics and compares them against others in their league, age group, or professional benchmarks.
   - Highlights underrated or overlooked talent that may have gone unnoticed.
4. **Predict Development Potential:**
   - Uses historical data and patterns to forecast how each player might grow and improve over time.
   - Helps identify players with the highest long-term potential.
5. **Provide Actionable Insights:**
   - Generates easy-to-read reports with rankings, strengths, weaknesses, and recommendations.
   - Provides insights and recommendations to support recruitment, training, and player development decisions.

Data Collection & Preparation:

Data is collected through:

- **Video Capture**: Record matches and training sessions in kasi communities using smartphones, tablets, or affordable cameras. Footage can be collected by **coaches**, **local volunteers**, **academy staff**, **or the players themselves**.

- **Metadata Collection**: Player information (name, age, position, height, and stats like goals, assists, minutes played) is recorded by **coaches**, **volunteers**, **or academy staff** to ensure accuracy.

- **Additional Data Sources**: Collect contextual and performance-related data from multiple sources, such as:
  - Player training logs and practice attendance
  - Coach or mentor observations on skills, attitude, and improvement areas
  - Local tournament results and match statistics
  - Partner academies or community programs contributing additional context

- **Labeling & Structuring**: Key actions like passes, dribbles, tackles, and shots are tagged. Videos are standardized, cleaned, and segmented into meaningful clips for ML analysis.



- **Scale & Reliability**: System designed to handle large volumes of data across multiple communities, similar to large-scale learning datasets (e.g., tens of thousands of players and millions of recorded interactions).

- **Cost-Effective & Scalable**: Leverages existing community resources; minimal equipment needed, and easily expandable across multiple locations.

- **Outcome**: Produces both visual and contextual data necessary for **performance evaluation and structured player insights**, while maintaining a structured, reliable, and scalable dataset.

Machine Learning

Type of ML :

Supervised learning with some unsupervised components.

- **Supervised**: To predict player potential or performance scores using labeled historical data (e.g., past players' metrics and outcomes).
- **Unsupervised** / clustering: To group players by style, strengths, or similarity when labeled outcomes are limited.



Data Preparation Considerations

- **Consistency**: Videos must have similar formats (resolution, frame rate, camera angle) to ensure reliable analysis.
- **Data Cleaning**: Remove low-quality, blurry, or incomplete footage to avoid incorrect model predictions.
- **Standardization**: Ensure player information (names, positions, IDs) is consistent across all data.
- **Segmentation**: Break videos into meaningful clips (e.g., passes, shots, dribbles) for accurate feature extraction.
- **Label Quality**: Ensure actions (passes, tackles, shots) are correctly tagged to train supervised models effectively.
- **Balancing Data**: Avoid bias by ensuring data includes different positions, age groups, and skill levels.

Algorithms and methodolgies

Computer Vision (Pose Estimation & Object Detection):

Used to extract player movements, ball interactions, and actions (passes, shots, dribbles) directly from video footage.

Feature Engineering:

Convert video data into measurable features such as speed, positioning, involvement, and decision-making patterns.

Supervised Learning (Regression / Classification):

- **Regression models** predict a player's potential score.
- **Classification models** are used internally to group players into potential categories (e.g., high, medium, low).

Gradient Boosting Models (e.g., decision-tree based):

Used for prediction due to their high accuracy on structured data and ability to handle different feature types..

Algorithms and methodolgies

Motivation:

- **Computer vision** removes the need for manual tracking and works with simple video input.
- **Supervised learning** allows the system to learn from labeled examples and make reliable predictions about player potential.
- **Gradient boosting models** are efficient, accurate, and perform well even with limited or imperfect data — making them suitable for real-world environments like kasi communities.
- The combination allows the system to turn **raw video into meaningful insights** and predictions, which is essential for identifying overlooked talent

Overfitting / Underfitting

- **Overfitting** is a likely challenge:
  - Due to limited and noisy data (e.g., inconsistent video quality, small number of players), the model may learn patterns that do not generalize well to new players.
- **Underfitting** may also occur initially:
  - If the model is too simple or the features extracted are not rich enough, it may fail to capture important aspects of player performance.
- **Mitigation strategies**:
  - Use more data as the system scales
  - Apply regularization techniques
  - Use cross-validation
  - Carefully engineer meaningful features from video

Model Evaluation & Performance



HOW WILL WE KNOW IT WORKS?

- **Train/Test Split:**
  - Split data into training and testing sets to evaluate how well the model generalizes to unseen players.
- **Performance Metrics:**
  - **Regression:** Mean Squared Error (MSE) for predicting player potential
  - **Classification:** Accuracy, Precision, Recall (for player categories)
- **Real-world validation:**
  - Compare model predictions with player performance trends over time as more data becomes available
  - Check if identified players show strong performance or progression over time
- **User feedback:**
  - Coaches and players validate whether insights and rankings are meaningful and accurate

The Rollout

How KasiScout Will Be Deployed:



Platform:

Built as a mobile app + web platform

- **Mobile app:** Used by players, coaches, and volunteers to record and upload videos and player information
- **Web platform**: Used internally to analyze player data and generate insights, which are then provided to scouts and clubs through curated recommendations

Rollout approach :

- Start with pilot communities (selected kasi areas)
- Test data collection and model performance
- Gradually expand to more regions



Partnerships & Real-World Integration:

- Work with local coaches and community teams for consistent data collection
- Partner with schools and academies to scale and structure data
- Collaborate with clubs and scouts who consult KasiScout for talent identification

User flow:

Record/upload match videos → System analyzes → Insights generated → KasiScout delivers recommendations and identified players to scouts/clubs

How KasiScout Generates Profit:



- **Consulting & Talent Recommendations**
  Used to extract player movements, ball interactions, and actions (passes, shots, dribbles) directly from video footage.
- **Subscription Model**
  Clubs subscribe for ongoing access to talent identification services and reports
- **Discovery / Success Fees**
  Fees charged when a player identified by KasiScout is successfully signed by a club



Value Proposition:

- Saves clubs time and scouting costs
- Provides access to hidden, high-potential talent
- Gives clubs a competitive advantage in recruitment

Responsible AI and Data Use



Data Privacy & Consent

- Collect data with player/guardian consent
- Securely store and protect personal data
- Allow users to opt out or delete data

Bias & Fairness

- Use diverse and representative data
- Monitor and reduce model bias
- Ensure fair evaluation despite data quality differences

Preventing Misuse

- Restrict access to sensitive insights
- Prevent exploitation of players
- Use the system to support development, not replace judgment

Transparency & Accountability

- Provide clear explanations of insights
- Keep a human in the loop for decisions
- Continuously monitor and improve the system

REVOLUTIONIZING TALENT DISCOVERY- ONE PLAYER AT A TIME

Football talent shouldn’t go unnoticed. It shouldn’t depend on being in the right place, playing in the right tournament, or having the right connections. Across kasi communities, thousands of players with raw potential remain unseen. With KasiScout, that gap is closed.

We use machine learning and video analysis to do what traditional scouting cannot: we don’t just observe players — we uncover hidden potential and transform raw performance into meaningful insights that help identify future stars.

Investing in KasiScout means investing in scalable, data-driven talent discovery — and unlocking a future where every player, no matter their background, has a real opportunity to be seen, developed, and succeed.

Kasi Scout.



The next generation of football stars won’t be found by chance , they’ll be found by KasiScout.

Thank You!